[INCH−POUND]
A−A−59590/16E
16 September 2025
SUPERSEDING
A−A−59590/16D
16 April 2020

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .375 X .375 INCH BODY SIZE, WEDGE MOUNTING BODY, WITH VISUAL LOCK INDICATION AND SCREW RETENTION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A−A−59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes herein).

AA59590	/16	B, C, H, R	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Configuration

Example: AA59590/16CG58 is the PIN for a chemical film finished, 5.8 inch (147.3 mm) long visual lock indicating card holder with a lock nut on the screw for captivation. The card holder also features three tapped mounting holes for use with metric M2.5 x 0.45 fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G-rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

Material. Unless otherwise specified herein, the card holder materials shall be as specified in A−A−59590.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, and 5, in table I, and A−A−59590.

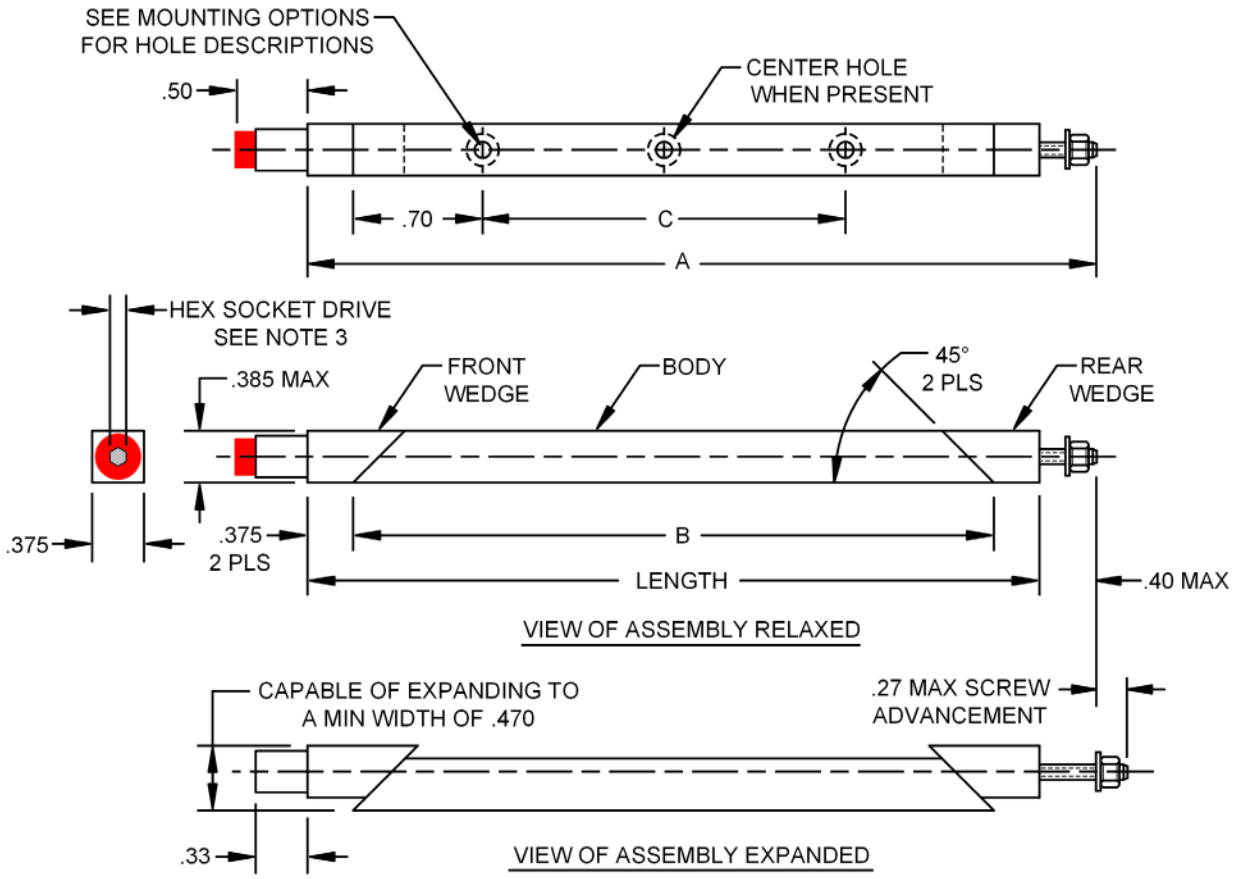
Visual lock indicator (see figure 2). Card holders shall have a visual indicator to show when the card holder is in its relaxed (unlocked) state. When the card holder is in the unlocked (relaxed) state, the actuating end of the screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

Screw retention (see figure 3). Card holders shall have a screw retention feature intended to prevent unintentional disassembly of the card holder. The locknut shall be fabricated of a corrosion resistant material.

Actuating screw hex drive socket. The across flats dimension for the hex drive socket shall be as follows: .140 inch (3.57 mm) for mounting options "J", "N", "T", "R", "V", and "S" and a metric dimension of 3.0 mm (0.118 inch) for mounting options "G" and "M".

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Inches	mm	Inches	mm	Inches	mm	Inches	mm
.27	6.9	.375	9.5	.40	10.2	.50	12.7
.33	8.3	.385	9.8	.470	11.9	.70	17.8

NOTES:

1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The across flats dimension for hex drive socket shall be .140 inch (3.57 mm) for mounting options "J", "N", "R", "T", "S", and "V". The across flats dimension for hex drive socket shall be a metric dimension of 3.0 mm (.118 inch) across flats for mounting option "G" and "M".

FIGURE 1. Relaxed and expanded dimensions.

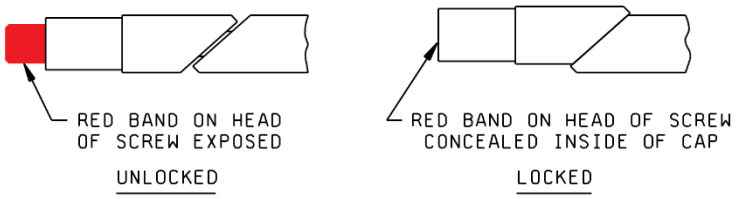


FIGURE 2. Visual lock indicator.

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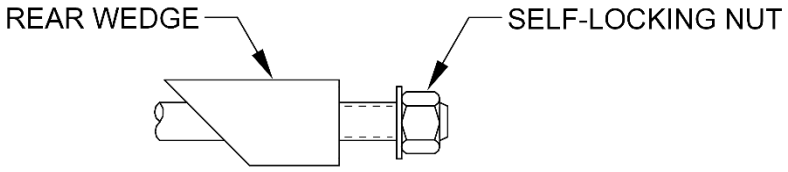


FIGURE 3. Screw retaining self-locking nut (corrosion resistant) details.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 16 to 20 inch−pounds (1.8 to 2.3 N−m) for assemblies using no options or option "W" and 17 to 21 inch−pounds (1.9 to 2.5 N−m) for assemblies using options "E" or "D".

Cold plate slot width dimensions (when applicable). The recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .425 inch (10.80 mm) plus the thickness of the printed board of the circuit card assembly (see A−A−59590).

Finish. The finish designator shall be as specified in A−A−59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "H" (hard black anodize), or "R" (clear chemical film).

Mounting. The mounting designators shall be as specified in A−A−59590. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "R" or "T" (tapped 0−80 UNF 2B holes), "V" or "S" (tapped 2−56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" may be shipped unassembled. See figure 1 and table I for mounting hole spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .136 inch (3.45 mm) diameter through holes, countersunk 100 degrees by .190 to .200 inch (4.83 to 5.08 mm) diameter with an access/clearance counterbore hole of .190 to .200 inch (4.83 to 5.08 mm) diameter by .300 inch (7.62 mm) deep.

Rivets. This card holder uses rivet style B as specified in A−A−59590 when rivet mounting is used.

Mounting hole locations (when specified). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole, when present, shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A−A−59590 and the available lengths are listed in table I.

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TABLE I. Assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ±.02 (0.5)	Dimension "A" (maximum)	Dimension "B" ±.03 (0.8)	Dimension "C" ±.005 (0.13)
28	2.8 (71.1)	3.45 (87.6)	2.05 (52.1)	.65 (16.5)
33	3.3 (83.8)	3.95 (100.3)	2.55 (64.8)	1.15 (29.2)
38	3.8 (96.5)	4.45 (113.0)	3.05 (77.5)	1.65 (41.9)
43	4.3 (109.2)	4.95 (125.7)	3.55 (90.2)	2.15 (54.6)
48	4.8 (121.9)	5.45 (138.4)	4.05 (102.9)	2.65 (67.3)
53	5.3 (134.6)	5.95 (151.1)	4.55 (115.6)	3.15 (80.0)
58	5.8 (147.3)	6.45 (163.8)	5.05 (128.3)	3.65 (92.7)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for general information only.

Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figures 1, 2 and 3, and may include those on figures 4 and 5. Card holders not requiring the options described by table II shall leave the configuration position in the PIN blank.

TABLE II. Configurations.

Configuration	Applicable figures	Hardware options
	1, 2, and 3	No added options
W	1, 2, 3 and 4	Lockwasher and flat washer
E	1, 2, 3 and 5	Screw self-locking element
D	1, 2, 3, 4 and 5	Lockwasher, flat washer and screw self-locking element

Lockwasher and flat washer (see figure 4). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening of the card holder assembly from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix "W" in the PIN (see classification and notes).

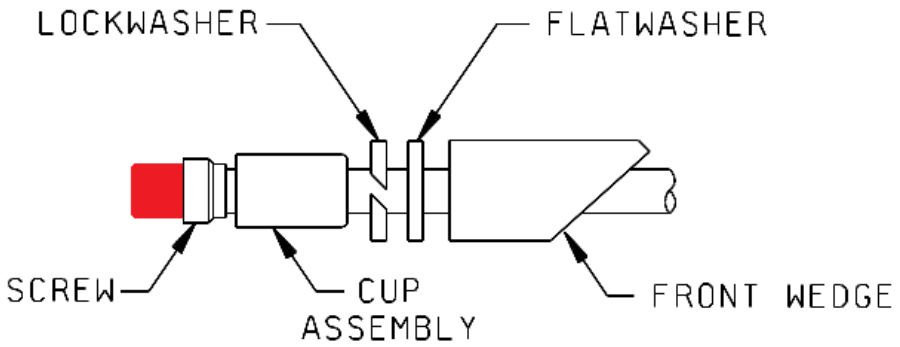


FIGURE 4. Lockwasher and flat washer details.

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Screw self-locking element (see figure 5). The use of a screw self-locking element will provide prevailing torque for resistance to loosening of the card holder assembly from shock and vibration. The screw self-locking element shall be as specified in A-A-59590. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).

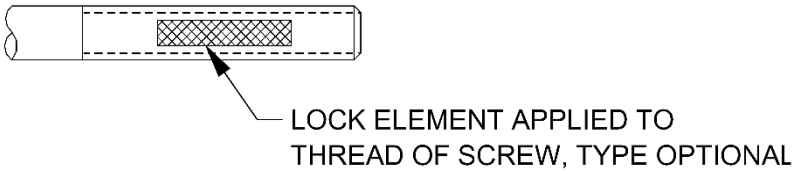


FIGURE 5. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix "D" in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A-A-59590 - Holder, Electrical Card, Wedge Retainers, 3 Piece, For Cold Plate Applications, General Requirements For.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Ordering data. Ordering data shall be as specified in A-A-59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
61081	Schroff, Inc. 7328 Trade Street San Diego, CA 92121-3410	Telephone: (858) 740-2400 Facsimile: (858) 740-2430 E-mail: askschroff@nVent.com URL: https://schroff.nvent.com
5BG68	American Circuit Card Retainers, Inc. 2310 E. Orangethorpe Avenue Anaheim, CA 92806-1231	Telephone: (714) 738-6194 Facsimile: (714) 446-0119 E-mail: sales@accrmfg.com URL: https://www.accrmfg.com

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Part number supersession data. These CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE III. Commercial part number supersession data.

PIN designator AA59590/16	Vendor similar designator or type part number 1/		
	CAGE 61081	CAGE 5BG68	
BJ28	VA250CR2.80HR	3370VBA-2.80HR	
BN28	VA250CR2.80R	3370VBA-2.80R	
BT28	VA250CR2.80T0R	3370VBA-2.80T0R	
BV28	VA250CR2.80T2R	3370VBA-2.80T2R	
BG28	MVA250CR2.80TM2.5R	3370MVBA-2.80TM2.5R	
BJ28E	VA250CR2.80HLR	3370VBA-2.80HLR	
BN28E	VA250CR2.80LR	3370VBA-2.80LR	
BT28E	VA250CR2.80T0LR	3370VBA-2.80T0LR	
BV28E	VA250CR2.80T2LR	3370VBA-2.80T2LR	
BG28E	MVA250CR2.80TM2.5LR	3370MVBA-2.80TM2.5LR	
BJ28W	WVA250CR2.80HR	3370VWBA-2.80HR	
BN28W	WVA250CR2.80R	3370VWBA-2.80R	
BT28W	WVA250CR2.80T0R	3370VWBA-2.80T0R	
BV28W	WVA250CR2.80T2R	3370VWBA-2.80T2R	
BG28W	MWVA250CR2.80TM2.5R	3370MVWBA-2.80TM2.5R	
BJ28D	WVA250CR2.80HLR	3370VWBA-2.80HLR	
BN28D	WVA250CR2.80LR	3370VWBA-2.80LR	
BT28D	WVA250CR2.80T0LR	3370VWBA-2.80T0LR	
BV28D	WVA250CR2.80T2LR	3370VWBA-2.80T2LR	
BG28D	MWVA250CR2.80TM2.5LR	3370MVWBA-2.80TM2.5LR	

See footnotes at end of table.

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TABLE III. Commercial part number supersession data - Continued.

PIN designator AA59590/16	Vendor similar designator or type part number 1/		
	CAGE 61081	CAGE 5BG68	
CJ28	V250CR2.80HR	3370VCG-2.80HR	
CN28	V250CR2.80R	3370VCG-2.80R	
CT28	V250CR2.80T0R	3370VCG-2.80T0R	
CV28	V250CR2.80T2R	3370VCG-2.80T2R	
CG28	MV250CR2.80TM2.5R	3370MVCG-2.80TM2.5R	
CJ28E	V250CR2.80HLR	3370VCG-2.80HLR	
CN28E	V250CR2.80LR	3370VCG-2.80LR	
CT28E	V250CR2.80T0LR	3370VCG-2.80T0LR	
CV28E	V250CR2.80T2LR	3370VCG-2.80T2LR	
CG28E	MV250CR2.80TM2.5LR	3370MVCG-2.80TM2.5LR	
CJ28W	WV250CR2.80HR	3370WVCG-2.80HR	
CN28W	WV250CR2.80R	3370WVCG-2.80R	
CT28W	WV250CR2.80T0R	3370WVCG-2.80T0R	
CV28W	WV250CR2.80T2R	3370WVCG-2.80T2R	
CG28W	MWV250CR2.80TM2.5R	3370MVWCG-2.80TM2.5R	
CJ28D	WV250CR2.80HLR	3370VWCG-2.80HLR	
CN28D	WV250CR2.80LR	3370VWCG-2.80LR	
CT28D	WV250CR2.80T0LR	3370VWCG-2.80T0LR	
CV28D	WV250CR2.80T2LR	3370VWCG-2.80T2LR	
CG28D	MWV250CR2.80TM2.5LR	3370MVWCG-2.80TM2.5LR	
HJ28	VHA250CR2.80HR	3370VBH-2.80HR	
HN28	VHA250CR2.80R	3370VBH-2.80R	
HT28	VHA250CR2.80T0R	3370VBH-2.80T0R	
HV28	VHA250CR2.80T2R	3370VBH-2.80T2R	
HG28	MVHA250CR2.80TM2.5R	3370MVBH-2.80TM2.5R	
HJ28E	VHA250CR2.80HLR	3370VBH-2.80HLR	
HN28E	VHA250CR2.80LR	3370VBH-2.80LR	
HT28E	VHA250CR2.80T0LR	3370VBH-2.80T0LR	
HV28E	VHA250CR2.80T2LR	3370VBH-2.80T2LR	
HG28E	MVHA250CR2.80TM2.5LR	3370MVBH-2.80TM2.5LR	

See footnotes at end of table.

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TABLE III. Commercial part number supersession data - Continued.

PIN designator AA59590/16	Vendor similar designator or type part number 1/		
	CAGE 61081	CAGE 5BG68	
HJ28W	WVHA250CR2.80HR	3370VWBH-2.80HR	
HN28W	WVHA250CR2.80R	3370VWBH-2.80R	
HT28W	WVHA250CR2.80T0R	3370VWBH-2.80T0R	
HV28W	WVHA250CR2.80T2R	3370VWBH-2.80T2R	
HG28W	MWVHA250CR2.80TM2.5R	3370MVWBH-2.80TM2.5R	
HJ28D	WVHA250CR2.80HLR	3370VWBH-2.80HLR	
HN28D	WVHA250CR2.80LR	3370VWBH-2.80LR	
HT28D	WVHA250CR2.80T0LR	3370VWBH-2.80T0LR	
HV28D	WVHA250CR2.80T2LR	3370VWBH-2.80T2LR	
HG28D	MWVHA250CR2.80TM2.5LR	3370MVWBH-2.80TM2.5LR	
RJ28	VR250CR2.80HR	3370VCC-2.80HR	
RN28	VR250CR2.80R	3370VCC-2.80R	
RT28	VR250CR2.80T0R	3370VCC-2.80T0R	
RV28	VR250CR2.80T2R	3370VCC-2.80T2R	
RG28	MVR250CR2.80TM2.5R	3370MVCC-2.80TM2.5R	
RJ28E	VR250CR2.80HLR	3370VCC-2.80HLR	
RN28E	VR250CR2.80LR	3370VCC-2.80LR	
RT28E	VR250CR2.80T0LR	3370VCC-2.80T0LR	
RV28E	VR250CR2.80T2LR	3370VCC-2.80T2LR	
RG28E	MVR250CR2.80TM2.5LR	3370MVCC-2.80TM2.5LR	
RJ28W	WVR250CR2.80HR	3370VWCC-2.80HR	
RN28W	WVR250CR2.80R	3370VWCC-2.80R	
RT28W	WVR250CR2.80T0R	3370VWCC-2.80T0R	
RV28W	WVR250CR2.80T2R	3370VWCC-2.80T2R	
RG28W	MWVR250CR2.80TM2.5R	3370MVWCC-2.80TM2.5R	

See footnotes at end of table.

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TABLE III. Commercial part number supersession data − Continued.

PIN designator AA59590/16	Vendor similar designator or type part number 1/		
	CAGE 61081	CAGE 5BG68	
RJ28D	WVR250CR2.80HLR	3370VWCC−2.80HLR	
RN28D	WVR250CR2.80LR	3370VWCC−2.80LR	
RT28D	WVR250CR2.80T0LR	3370VWCC−2.80T0LR	
RV28D	WVR250CR2.80T2LR	3370VWCC−2.80T2LR	
RG28D	MWVR250CR2.80TM2.5LR	3370MVWCC−2.80TM2.5LR	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A−A−59590.

PIN length examples. The CID PINs listed in table IV are for all available standard card holder lengths for this specification sheet. However, only one specific finish, mounting and configuration are listed (see PIN example for a break-down of the codes).

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/16	Vendor similar designator or type part number 1/ 2/		
	CAGE 61081	CAGE 5BG68	
CG28D	MWV250CR2.80TM2.5LR	3370MVWCG−2.80TM2.5LR	
CG33D	MWV250CR3.30TM2.5LR	3370MVWCG−3.30TM2.5LR	
CG38D	MWV250CR3.80TM2.5LR	3370MVWCG−3.80TM2.5LR	
CG43D	MWV250CR4.30TM2.5LR	3370MVWCG−4.30TM2.5LR	
CG48D	MWV250CR4.80TM2.5LR	3370MVWCG−4.80TM2.5LR	
CM53D	MWV250CR5.30ETM2.5LR	3370MVWCG−5.30ETM2.5LR	
CM58D	MWV250CR5.80ETM2.5LR	3370MVWCG−5.80ETM2.5LR	

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A−A−59590.
2/ Other lengths are available on request.

Guidance on use of alternative parts with less hazardous or non–hazardous materials. This CID specification sheet provides for a number of alternative plating materials via the PIN. Users should select the PIN with the least hazardous material that meets the form, fit, and function requirements of their application.

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Subject term (key word) listing.

Actuating screw
Cold plate
Counterbore holes
Expanded dimension
Installation torque
Mounting holes
Relaxed dimension
Self locking nut
Visual lock indicator

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

MILITARY INTERESTS:

Custodians:
Army − CR
Navy − EC
Air Force − 85
DLA − CC

CIVIL AGENCY COORDINATING ACTIVITY:

GSA − FAS

Preparing Activity:
DLA − CC

Project 5998−2025−008

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